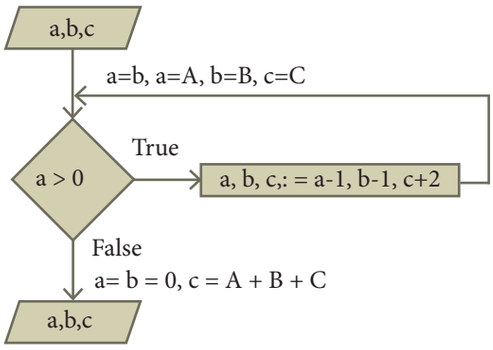

Figure 7.7: Algorithm monochromatize

7.3 Decomposition

Problem decomposition is one of the elementary problem-solving techniques. It involves breaking down a problem into smaller and more manageable problems and combining the solutions of the smaller problems to solve the original problem. Often, problems have structure. We can exploit the structure of the problem and break it into smaller problems. Then, the smaller problems can be further broken until they become sufficiently small to be solved by other simpler means. Their solutions are then combined together to construct a solution to the original problem.

7.3.1 Refinement

After decomposing a problem into smaller subproblems, the next step is either to refine the subproblem or to abstract the subproblem.

1. Each subproblem can be expanded into more detailed steps. Each step can be further expanded to still finer steps and so on. This is known as refinement.
2. We can also abstract the subproblem. We specify each subproblem by its input property and the input-output relation. While solving the main problem, we only need to know the specification of the subproblems. We do not need to know how the subproblems are solved.

Example 7.6. Consider a school goer's action in the morning. The action can be written as

1 Get ready for school

We can decompose this action into smaller, more manageable action steps which she takes in sequence:

1 Eat breakfast

2 Put on clothes

3 Leave home

We have refined one action into a detailed sequence of actions. However, each of these actions can be expanded into a sequence of actions at a more detailed level and this expansion can be repeated. The action "Eat breakfast" can be expanded as

1 -- Eat breakfast

2 Eat idlis

3 Eat eggs

4 Eat bananas

The action "Put on clothes" can be expanded as

1 -- Put on clothes

2 Put on blue dress

3 Put on socks and shoes

4 Wear ID card

and "Leave home" expanded as

1 -- Leave home

2 Take the bicycle out

3 Ride the bicycle away

Thus, the entire action of "Get ready for school" has been refined as

1 -- Eat breakfast

2 Eat idlis

3 Eat eggs

4 Eat bananas

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